

Ansh Sharma

Backend & AI Engineer · FastAPI · LangGraph · Distributed Systems · RAG · pgvector

Built and shipped multi-agent AI systems serving real users

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Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar
Bachelor of Technology (B.Tech)

July 2022 – June 2026

Experience

Siemens – *Technical Intern (Software & Logic Automation)*

June 2025 – August 2025

- Designed a **closed-loop tank automation system** in Siemens Ladder Logic – LLS/HLS sensors trigger motor and drainage state transitions; system runs continuously, functionally equivalent to a **two-interrupt state machine**.
- Modeled sequential control logic for **SCADA-based** workflows as explicit State Machines – made execution flow auditable across physical actuator states without a debugger; studied Siemens PLC/HMI stack and how **real-time control software** maps to factory-scale physical systems.

Projects

LinkedOut – Multi-User AI Outreach Automation System

[GitHub](#) | [Live Demo](#)

FastAPI · LangGraph · Playwright · PostgreSQL + pgvector · AWS EC2/S3 · Supabase · Next.js · OAuth2/JWT

- Architected and deployed** a multi-user distributed pipeline – scrapes LinkedIn jobs, scores them via **LLM inference**, finds alumni/HR contacts, generates **vector-embedding-based outreach notes**, and sends connection requests end-to-end.
- Fixed a multi-tenant concurrency bug**: replaced a shared global SQLAlchemy session (cross-user data leaks under load) with per-request isolated sessions via FastAPI's `yield` dependency – standard **distributed systems** isolation applied to a multi-tenant REST API.
- Chose **LangGraph over LangChain** for stateful agentic execution – used `interrupt()` with `MemorySaver` checkpointing to pause the pipeline at two human-approval steps and resume with full state (resume JSON, scored jobs, notes) intact.
- Diagnosed **LinkedIn's silent bot detection** (EC2 IPs returning 0 results, no error) – root-caused to browser fingerprinting; fixed with `playwright-stealth` + randomized delays + cookie reuse; added **client-side rate limiting** at 15 req/day to prevent bans.
- Designed **multi-tenant auth**: OAuth2/JWT, all DB tables scoped by `user_id` FK, S3 paths and browser cookies namespaced per user – zero data crossover under concurrent sessions.

Agentic Document QA System – RAG Pipeline with LLM Inference

[GitHub](#) | [Live Demo](#)

FastAPI · React · LangGraph · FAISS · Sentence Transformers · Groq/Llama 3.2 · Vector Embeddings

- Built a **production RAG pipeline** over 500+ pages – embedded docs into a **FAISS vector index**; benchmarked semantic vs. fixed-size chunking – semantic chunking improved retrieval relevance on ambiguous queries, validated on 50 manual test cases.
- Added a **pre-LLM validation gate** to filter low-relevance chunks before inference, reducing hallucinations; used **Groq-hosted Llama 3.2** for sub-second **LLM inference** via async FastAPI.

Real-Time Image Classification API – CV Model Serving System

[GitHub](#)

Flask · PyTorch · EfficientNet-B3 · Transfer Learning · REST API · Response Caching

- Fine-tuned EfficientNet-B3** via PyTorch transfer learning specifically for food classification – identifies the food item from an uploaded image and triggers a **two-stage pipeline**: CV inference → Edamam API call to fetch macros (calories, protein, fat) → rendered nutrition breakdown for the user.
- Deployed as a **Flask REST API** with a **JSON response cache** (decorator pattern on route layer) – eliminated redundant Edamam API calls on repeated queries and cut p50 response latency by **40%**.

Technologies

Languages: Python, C++, JavaScript, SQL

Backend & System Design: FastAPI, Flask, Node.js, REST APIs, Microservices, WebSockets, asyncio, OAuth2, JWT, Rate Limiting, Async Task Queues, CI/CD

AI/ML & LLM Systems: LangGraph, LangChain, RAG Pipelines, Vector Embeddings, Multi-Agent Systems, Prompt Engineering, LLM Inference, PyTorch, Hugging Face, Sentence Transformers, Groq API, Llama 3.2

Databases & Vector DBs: PostgreSQL, pgvector, FAISS, Redis, Supabase

Cloud & DevOps: AWS (EC2, S3), Docker, Nginx, Certbot (SSL), GitHub Actions (CI/CD), Vercel

Frontend & Tools: React, Next.js, Tailwind CSS, Playwright, Git, Postman, Jupyter

Achievements

- Global rank ~4,000** in a LeetCode Biweekly Contest.
- Solved **200+ DSA problems** across LeetCode, Coding Ninjas, and GeeksforGeeks.
- Core Member**, LADC Debating Society – competitive academic debate at NIT Jalandhar.